

Vector Calculator

3D dot · cross · real-time visualization

Compute the dot and cross product of two 3D vectors step by step, and visualize the result in 3D space.

Cross product (A × B)

Dot product (A · B)

Vector A

x

1

y

2

z

0

Vector B

x

0

y

1

z

10

Magnitude

|A| = 2.2361

|B| = 10.0499

|A×B| = 22.383

(parallelogram area)

Result

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} 20 \\ -10 \\ 1 \end{pmatrix}$$

Working

Step 1: Vectors

$$\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 0 \\ 1 \\ 10 \end{pmatrix}$$

Step 2: Cross product formula (determinant expansion)

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 0 \\ 0 & 1 & 10 \end{vmatrix}$$

Step 3: Compute each component

$$i : (2)(10) - (0)(1) = 20$$
$$j : (0)(0) - (1)(10) = -10$$
$$k : (1)(1) - (2)(0) = 1$$

Result

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} 20 \\ -10 \\ 1 \end{pmatrix}, \quad |\mathbf{a} \times \mathbf{b}| = 22.383$$

3D visualization Drag to rotate

Vector A

Vector B

Vector A×B

